Prospectus: Using phylogenies to understand and predict physiological adaptations to anthropogenically induced challenges

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Global change, including rising temperatures and the spread of zoonotic diseases, has forced species to adapt to keep up or perish. Organisms deal with a changing environment by either genetic adaptation involving changes in allele frequencies or through phenotypic plasticity. Temperature affects many body processes through its effect on protein functioning, ranging from respiration, reproduction, locomotion, digestion, and metabolism. Consequently, genetic adaptation might come at a high cost. On the other hand, genetic variation in heat tolerance (even if small) could result in physiological diversification over macroevolutionary time. Understanding species' capacity to respond physiologically to new ecological challenges is essential to understanding their constraints and limitations to evolve and adapt to changing environments.

How species have evolved in the past might provide important information regarding specie's evolutionary potential. For example, endotherms evolved homeothermy millions of years ago, opening new niches and opportunities that may not have previously existed (Nespolo et al., 2011). Particularly, mammals and birds occupy wider ranges than ectotherms and have evolved across environmental gradients faster (Rolland et al. 2018). Thermal physiology is also a mechanism to fight off pathogens that would otherwise be detrimental. Because of this, fungal infections have rarely been much of a concern in endotherms (Casadevall, 2005). Even though evolution leaves a clear signal on species' physiological and ecological diversification, most studies on a phylogenetic scale have examined morphological traits and ignored important physiological characteristics that are likely tied to the environment. Studies that integrate physiological, behavioral, and morphological characters with phylogenetic approaches could help the development of better predictions of species adaptations. The phylogenetic history covers species' adaptations to a changing world, and future studies should work on incorporating species relatedness and the sum of adaptations and acclimation that a species acquired and genetically passed on to the next generation. The need to connect and accurately predict the rate of species adaptations to anthropogenically induced challenges will give managers and policymakers a better understanding of the spread of infectious diseases, better predict outbreaks, and provide better solutions and preventive procedures for its limited resources.

Macroevolutionary predictability of mammal responses to novel physiological threats is an understudied phenomenon that my research seeks to address. Here, I aim to **1) compare the adaptations and tolerance to climatic conditions in endotherms and ectotherms** and **2) predict species susceptibility to the spreading fungal disease in bats (White-nose syndrome) caused by the fungus *Pseudogyymnoascus destructans* across the western United States.** Chapter 1 will evaluate how body temperature has evolved in endotherms and ectotherms. Most

comparisons of body temperature of endotherms and ectotherms have accounted for past and present climate conditions but lacked the proper statistical models that allow incorporating thermal physiology and environmental variables in a phylogenetic framework. Chapter 2 will address the gap in knowledge relating to the phylogenetic predictability of susceptibility to pathogenic diseases in New World bats and species' susceptibility to the spreading fungal infection in bats (White-nose syndrome) caused by the fungus *Pseudogyymnoascus destructans* across the western United States. This information will aid in accurately predicting and mitigating measures to conserve these species.

Chapter 1. A comparison of body temperature inertia within and among the major clades of endotherms and ectotherms

Background

Body temperature (Tb) has evolved differently across clades, especially endotherms and ectotherms. The term 'hotter is better' and 'warmer is better' has been widely argued to support that animals naturally thrive in warmer temperatures due to increased metabolic reactions (Angilletta et al., 2010). However, considerable stasis in thermal physiological traits has been observed (Moreira et al., 2021). This is likely due to biochemical constraints in the upper limit and fewer constraints to protect the lower boundary from the cold climates (Muñoz et al., 2014). Asymmetry in the response at lower temperatures and not at the higher temperature may put some species at risk of increasing temperatures (Garcia-Porta et al., 2019). Conversely, the fine range of differences in temperature tolerance in individuals could get passed to the next generation and eventually be the species adapted temperature.

Previous phylogenetic studies have found differences between endotherms and ectotherms, primarily in evolutionary rate and relationships of body temperature to the environment. Quantifying species adaptation thresholds to increasing temperatures should integrate models that capture the complexity of the species' behavioral, physiological, and evolutionary histories. Still, studies have failed to consider all of these simultaneously so far. How differently do endotherms and ectotherms respond to their environment, and how does this impact their response to climate change? Moreira et al. (2021) observed distinct body temperature evolution in both endotherms and ectotherms and among the major clades. Mammals and birds had the lowest rates of 0.084 and 0.078 $℃^2$ per million years (my), respectively, while amphibians and crocodiles had higher rates of 0.165 and 0.116 $℃^2$ per my, respectively. Qu & Wiens (2020) observed weak correlation between climatic niche variables and physiological parameters in squamates(r^2=0.03), birds((r^2<0.01), mammals((r^2=0.02) and observed the strongest correlation in amphibians (r^2=0.25). However, they only had the environmental temperature in their analysis. Importantly, they didn't account for the heterogeneity of the niches from many other factors known to be important from other studies, including precipitation and behavioral and

physiological traits (see below). Rolland et al. (2018) tested for tropical niche conservatism and found the highest rates of niche evolution in endotherms than ectotherms, which is the reverse of the pattern seen in body temperature evolutionary rates.

While a large variety of studies have examined body temperature evolution; none of the previous analyses have considered "phylogenetic inertia" or slowness of adaptation when measuring relationships between Tb and climatic variables when comparing endotherms and ectotherms (Labra et al., 2009, Hansen & Orzack, 2005, Hansen et al., 2008). To remedy this, I will repeat analyses using SLOUCH (Hansen et al., 2008) to analyze how strongly body temperature responds to climate after including all these previously studied correlates. **From this, I hypothesize that endotherms will respond more weakly to environmental temperatures, with a wide range of values allowed (like from higher latitudes) and that most ectotherms will evolve strongly toward specific climate optima but with significant phylogenetic inertia that may be compared between groups after accounting for other factors.** We propose integrating all of these studies by combining environmental, physiological, and phylogenetics to test whether endotherms and ectotherms have similar evolutionary rates and phylogenetic signals/inertia after accounting for measurable factors.

Differences in rate and responsiveness may exist because of behavioral differences in thermoregulation. Because adapting to ever-increasing temperatures during the day is more costly than its potential benefits (Tattersall et al., 2012), they must rely on behavioral adaptations. These behaviors buffer organisms well during the day but less during cooler nights (Bogert effect) (Domínguez–Guerrero et al., 2019). This potentially creates a constraint on how high their body temperature could get. Endotherms regulate their body temperature without much influence on their environments. It will be interesting to see if this phenomenon will be held while comparing two different times for nocturnal and diurnal species. **We hypothesize that there will be a statistical difference in body temperature between diurnal and nocturnal species in endotherms and more so in ectotherms, and so considering diel activity, hibernation, and other so-called "sleep-and-hide" traits in our analysis may provide a clearer picture of the impact of environmental temperature on thermal physiology.**

Additionally, unique physiological traits such as rumination impact some mammals' thermal preferences. Artiodactyls have evolved rumination for more efficient digestion. This process produces heat from rumination movements and metabolic activities, which greatly raises body temperature (Ferreira et al., 2021). One theory for coping with this is the network of arteries, *i.e.,* rete mirabile in artiodactyls that weaves around the nasal cavity before taking blood into the brain. This adaptation helps them cool the blood flowing to the brain when temperatures rise too high (Fukuta et al., 2007). Nonetheless, (Tattersall et al., 2012) tested the temperatures of Wildebeest in strenuous activities and observed a stronger link to prevention of water loss than to

an increase in temperature. **This leads us to hypothesize that Tb will most closely correspond to precipitation rather than temperature in this clade. To our knowledge, this study has not been done before.**

After accounting for all of these factors jointly, to what extent is there a distinction in how this physiological parameter evolves between groups? How much phylogenetic signal and inertia remains to compare the rates of Tb evolution among endotherms and ectotherms after accounting for known factors in major clades? **Here we will compare Tb in Endotherms and Ectotherms, among the major clades, and in Diurnal and Nocturnal species. We will further compute and compare their rates of evolution with environmental variables to provide insight into which will be more at risk with the rising temperatures.**

Data and methods

I will incorporate phylogeny and environmental temperatures and behavior to see if they influence Tb. Environmental variables, phylogeny, and behavior could be phylogenetically conserved, as observed in some lizard lineages (Grigg & Buckley, 2013). Environmental variables include; Mean annual Temperature (Ta) and Mean annual precipitation. It has been recommended that while looking for a species adaptation, not only focus on mean temperatures but also on other bioclimatic variables (Williams et al., 2012). I will fit models with phylogenetic regressions using the software *SLOUCH* (Hansen et al. 2008), which models the evolution of trait (in this case, Tb) in response to randomly evolving predictors (like Ta) while simultaneously estimating phylogenetic inertia or slowness due to adaptation. This provides an estimate of the "optimal" regression (in addition to the evolutionary regression) which is defined as the relationship between variables that would exist given an infinite amount of time for adaptation in a stable environment.

To test the hypothesis that behavioral thermoregulation is the cause of the most stasis in Tb and not physiological adaptations, I will analyze behavioral activities that will include nocturnal, diurnal, crepuscular, and arrhythmic (diel) preferences. These activity patterns could potentially vary throughout the day to influence Tb. (Crompton et al., 1978) speculated that the evolution of homeothermy started with a shift of mammals to nocturnal habitats from diurnal habitats as they were largely uninhabited. This would, in turn, lead to a lower than average body temperature for an equal-sized diurnal animal. I**n my analysis, I aim to see if these behaviors influence Tb and to focus on comparing the rates of Tb evolution between endotherms and ectotherms** to see if there is any difference between the two groups.

Tb data

We acquired a squamate thermo-physiology dataset from (Munoz et al., unpublished) and (Moreira et al., 2021), while the mammal, bird, and additional Squamata data were downloaded

from the Species 360 website's normal reference values from veterinary studies (Species360, http://zims.Species360.org). All datasets contained values for body temperature, and many include standard error values for these measurements. We replaced all missing standard errors with the median standard error value for body temperature in each dataset to account for any missing standard error values. We then merged mammal and squamate datasets with their respective data frames containing the extracted WorldClim values for only annual mean temperature and precipitation.

Ecological Data Collection.

All species occurrences for squamates and mammals were requested and downloaded from the Global Biodiversity Information Facility (GBIF, https://www.gbif.org/). Species occurrences were then imported into the programming software, R (R Core Team 2021), using the function "occ_download_get" from the package *rgbif* (Chamberlain et al. 2021). After all species occurrences were imported, we subsetted for species' scientific names, longitudes, latitudes, species names, and each species country code using the "select" function from the package, *dplyr* (Wickham et al. 2021). We filtered out any missing values for latitude and longitude from all species occurrences using the function "filter," also from the *dplyr* (Wickham et al. 2021) package. We then cross-checked and cleaned all species occurrences using the function "clean_coordinates" from the package *CoordinateCleaner* (Zizka et al., 2019). 3,621,382 squamate species occurrences and 7,481,576 mammalian species occurrences were filtered from the original requested number of occurrences from GBIF for squamates and mammals, respectively.

Climate data

Using the occurrences obtained above, climatic variables from the online database, WorldClim version 2 (http://www.worldclim.org), were extracted for each species occurrence using the function "extract" from the package, *raster* (Hijmans 2021). Out of the 19 climatic variables, only annual mean temperature (BIO1) and annual precipitation (BIO12) values were subsetted and used for our comparative phylogenetic analyses. The median mean annual temperature and annual precipitation values were summarized for each species for statistical purposes. We decided to use climate data from our GBIF occurrences instead of climate data from the exact locations where the body temperature data was collected because we felt that the latter would most likely strengthen our results.

Diel activity data.

Diel activity data used was by (Moreira et al., 2021), where I obtained data for 501 ectotherms and 857 endotherms. However, there is no single way of grouping the species according to their DIEL activity. I added nocturnal and diurnal; many fall under crepuscular and arrhythmic. This

might not be accurate for each species and might produce a bias. To account for this probable error, I will first analyze only the nocturnal and diurnal species per se and add the species I have additionally grouped and compare the difference.

Rumination & Aquatic habit

With the exception of two primitive artiodactyla, All artiodactyls are ruminants (Fukuta et al., 2007). I will analyze and compare the temperature evolution of this clade with the rest of the endothermic species. Additionally, I will code taxa as either predominantly aquatic or terrestrial and test whether these also predict significant variation in thermal traits.

Preliminary Results

I have already conducted several analyses; however, I still need to conduct a final analysis with all predictors synthesized from recently published papers (Qu & Wiens, 2020, Moreira et al., 2021). Our preliminary analyses included body temperature data from 188 species of squamates and 807 species of mammals. We tested the relationships between body temperature and our climatic variables, annual mean temperature, and annual precipitation using the SLOUCH package (Hansen et al. 2008). Based on our SLOUCH models, we found a statistically significant, positive relationship between body temperature (T_b) and environmental temperature for both optimal (Slope ± SE = 0.567 ± 0.150; R^2 = 10.6%; Fig. 1A; Table 1) and evolutionary regressions (Slope ± SE = 0.254 ± 0.0677; Fig. 1A; Table 1) in squamates. The optimal regression for squamates had a steeper and stronger relationship between annual mean temperature and body temperature than the evolutionary regression, which was reported to be quite weak in previous studies (Qu & Wiens, 2020). We also found a statistically significant, negative relationship between body temperature and precipitation for both optimal (Slope ± SE = -0.00312 ± 0.000841; Fig. 1B; Table 1) and evolutionary (Slope ± SE = -0.0014 ± 0.000377; Fig. 1B; Table 1) regressions within squamates. These relationships are much stronger than in previous studies due to our inclusion of phylogenetic lag and synthesizing across studies with additional predictors.

Likewise, the optimal regression for the relationship between annual precipitation and body temperature had a steeper slope than the evolutionary regression. In contrast, mammals exhibited a statistically significant, very weak, negative relationship between annual mean temperature and body temperature for both optimal (Slope ± SE = -0.0336 ± 0.00904; R^2 = 1.64%; Fig. 1C; Table 1) and evolutionary (Slope ± SE = -0.0203 ± 0.00549; Fig. 1C; Table 1) regressions. The optimal regression for annual mean temperature and body temperature in mammals was slightly steeper than the evolutionary regression. Mammals also showed little to no relationship between annual precipitation and body temperature for both optimal (Slope ± SE = 0.0000679 ± 0.0000928; Fig. 1D; Table 1) and evolutionary regressions (Slope ± SE = 0.0000412 ± 0.0000563; Fig. 1D; Table 1). The optimal and evolutionary regressions for annual precipitation and body temperature were close to zero. We will repeat our analyses by including diel activity, aquatic habit, and other

described variables. Importantly, after accounting for predictors and the varying responsiveness to environmental change, squamates and mammals showed similar phylogenetic inertia or slowness in physiological trait evolution (as measured by half-life, Table 1), indicating similar constraints on the evolution of thermal physiology.

TABLES AND FIGURES

Table 1. Comparison of SLOUCH models for the response trait, body temperature (ºC), in squamates and mammals.

						Optimal		Evolutionary			
Species Group	Predictor(s)	$t_{1/2}$ (mya)	α (mya^{-1})	σ^2 (ºC/mya)	ν_y (ºC^2)	Intercept (±SE)	Slope (±SE)	Intercept (±SE)	Slope (±SE)	R^2	AICc
Squamates	Temp	84.3	0.008	0.34	20.9	26.34 ± 1.66	0.42 ± 0.14	26.34 ± 1.66	0.19 ± 0.067	0.043	972
Squamates	Precip	59.9	0.011	0.39	17.0	31.54 ± 0.92	-0.0018 ± 0.00067	31.54 ± 0.92	-0.0010 ± 0.00037	0.037	972
Squamates	**Temp + Precip**	**87.4**	**0.007**	**0.26**	**16.8**	**26.80 ± 1.62**	**0.56 ± 0.15**	**26.60 ± 1.62**	**0.25 ± 0.067**	**0.106**	**963**
							-0.0031 ± 0.00084		**-0.0014 ± 0.00037**		
Mammals	Temp	31.6	0.021	0.14	3.4	35.91 ± 0.30	-0.029 ± 0.0069	35.91 ± 0.30	-0.023 ± 0.0054	0.022	2660
Mammals	Precip	32.1	0.021	0.15	3.5	35.59 ± 0.29	-0.00011 ± -0.000071	35.59 ± 0.29	-0.000089 ± 0.000056	0.003	2670
Mammals	**Temp + Precip**	**65.3**	**0.010**	**0.07**	**3.6**	**35.20 ± 0.58**	**-0.033 ± 0.0090**	**35.20 ± 0.58**	**-0.020 ± 0.0054**	**0.016**	**2550**
							0.000067 ± 0.000092		**0.000041 ± 0.000056**		

Figure 1. Optimal and evolutionary regressions for annual mean temperature **(A)** and annual precipitation **(B)** against body temperature in squamates for the best-fitting model. The optimal and evolutionary regressions of annual mean temperature **(C)** and annual precipitation **(D)** against body temperature in mammals. Evolutionary regressions are represented by the dashed, blue lines, and optimal regressions are represented by the solid, green lines.

Predictor variables included mean annual temperature (^{O}C) and annual precipitation (mm) based on WorldClim data extracted from our GBIF species occurrences. Phylogenetic half-life ($t_{1/2}$) values are millions of years based on the tree lengths for our squamate (228 mya) and mammal (mya) phylogenies. Rate of adaptation (α) values are in mya^{-1}, stationary variance (ν_y) values are in $^{o}C^2$, and evolutionary rate (σ^2) values are in $^{o}C^2$/mya. The slope (±SE) and intercept (±SE) of the optimal and evolutionary regressions for squamates and mammals were corrected for measurement error in the body temperature data collected. The values in bold were the chosen models based on AICc values. Note that all the best-fitting models are comprised of both temperature and precipitation environmental variables.

Chapter 2. How do the evolutionary contingencies of bats affect their response to white-nose syndrome?

Background Information

The fungus *Pseudogyymnoascus destructans* that causes white-nose syndrome has been present in Eurasia for thousands of years. It has only been recently observed in North America within the last 16 years. Due to this, bats in Eurasia are more resistant to the fungus, with reduced infections observed (endemic). In contrast, bats in North America, where the pathogen is introduced, suffer high mortality compared to bats in Eurasia. *P. destructans* can infect and kill multiple species of bats in North America. However, there is evidence that it doesn't kill with the same intensity in each bat species. Differences in mortality among bat species have been observed within the same hibernacula due to their corresponding differences in body size, hibernating temperature preferences, and metabolic reactions (Hayman et al., 2016).

Among other factors, there is evidence this variation is related to the thermal experiences of each species. Hibernating bats don't go into torpor the whole hibernating period. They periodically rise to their normal body temperatures (Tb) every few weeks for 1-3 hours (Hoyt et al., 2021). These are periods of the highest metabolic activities, and though it is only 1% of the total hibernation period, it accounts for the use of 80-90% of their fat reserves (Reeder et al., 2012, Hoyt et al., 2021). The fungal infections cause bats in torpor to "wake" more frequently than 'normal,' which leads to their death from the depletion of fat to sustain them through their hibernation period. It is unclear why they do this, but the trend has been consistent across all hibernating animals. Sometimes, they (could) shift to a different location within the same hibernacula and transfer pathogens (Hoyt et al., 2021). Further, these different spots and crevices have different microclimates, and different species have different preferences.

Most bat species prefer temperatures between 2^0C-10^0C during hibernation (Johnson et al., n.d.). Negative temperatures will be too low, so it will be energetically expensive to maintain a stable temperature for basic metabolic activities. Higher temperatures will make them spend most of their fat reserves before the end of winter. While the fungus thrives at temperatures higher than

6^0C. They will cease growing at greater than 19.7^0C and below 0^0C, providing a temperature preference overlap for infection. However, there has been a change in colder preference since the onset of WNS in the United States (Johnson et al., n.d.).

Studies seeking to understand the correlations of fungal loads in bat hibernacula and if species with high loads of WNS share their life histories have found they are phylogenetically related (Zukal et al., 2014). **Using a more recent fungal load data set (Dr. Hoyt, Unpublished), I would like to test for this signal and further predict the susceptibility of species not yet infected.** The predictive ability of phylogeny comes from its ability to "fill in the gaps" of unmeasured or unobserved traits that influence species susceptibility by using related species responses. Thus, combining phylogeny with measured variables can predict species responses and evaluate associations with measured variables. Variables I will include are: 1) Cluster size because it is species and temperature-dependent 2) Body size (in mass and forearm length) as this may correspond with fat reserves and thermal buffering and hence its capacity to survive the fungus 3) habitat type and 4) exposure to the fungus. Integrating these in a phylogenetic analysis will better predict what plays the most significant role in their spread, knowledge about what species are most infected/impacted, and potentially increase the model's predictive capacity for species yet to be exposed.

Proposed Methods

Data and phylogeny

Data contain fungal load data of 30 species from North America and Eurasia and their corresponding temperatures. Data will also include their forearm length, cluster size, and body size. Using the R package Ape, I pruned the tree first to contain the 30 species of bats in Eurasia and North America with the available fungal load data for the preliminary analysis. In my later research, I will include the species of bats in the Western United States to be predicted.

Phylogenetic linear models,

The phylogenetic linear model comes from the packag**e** R package *phylopars* (Bruggeman et al., 2009). This model first matches the tip labels of the phylo tree with the data names, then the tree traits, and fits an OU model to compute the phylogenetic signal.

Preliminary results

We did a phylogenetic linear regression model on these variables to estimate the ones that will be the most correlated to fungal loads observed in the group of 30 bat species with known fungal load data. Before analyzing, we log-transformed the variables to account for outliers that may try to 'pull' the prediction. We tested for the correlation of each variable to the fungal loads, and they are all significantly correlated. We conducted a series of phylogenetic linear regressions using a large number of combinations of variables using the R package *Phylopars,* which allows

the inclusion of missing data that can be phylogenetically imputed. We performed model selection using a combination of R^2 and variable significance (imputed data complicate model selection using other metrics). The model *log fungal loads ~ mean mass + forearm length + cluster size + temperature* had the most statistically significant relationship to the fungal loads, with an R^2= 0.39. For comparison, we did a Non-phylogenetic linear regression that does not make any assumptions about its ancestral history. This analysis suggested a weaker correlation with the fungal load with R^2 0.33. We will now compare which of these models performs better in cross-validation (see below). From the model, it's clear that several variables are highly correlated; for example, forearm length and mean mass both measure body size, and the *touching* may partially account for cluster size. Another complication is the confounding of geography and the epizootic/endemic status of the fungus.

Cross-validating

Cross-validation is the process in which the model is first tested for its validity by training it with the available data. Closely related species usually have similar traits to those distantly related. Using the 30 species with available data, we performed a 'leave one out cross-validation' by removing each species one at a time and calculating the expected fungal-load value for the removed species. The best model will have the lowest difference between the predicted and estimated values. The best-fitted preliminary model above produced predicted values positively correlated with the true values (Figure 3). I would like to use the model to predict the fungal loads of species not yet studied that occur further west of the United States Figure 4.

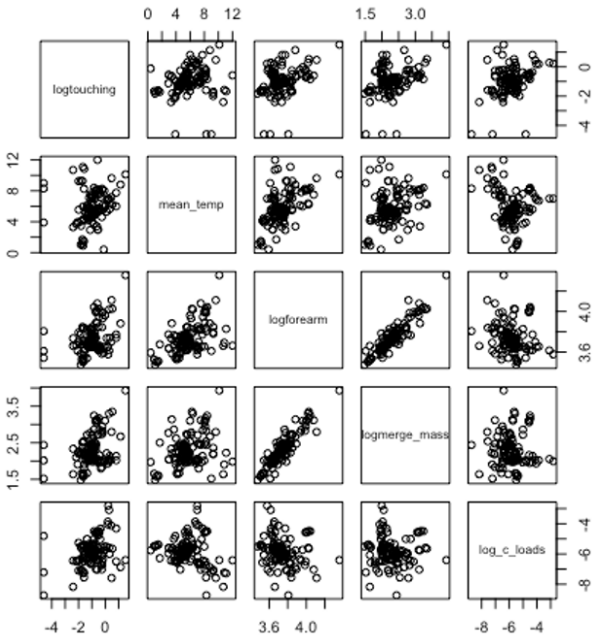

Figure 2; Correlations between the variables in the model i.e temperature, forearm length, mass, and Fungal load. Note; that all variables individually show statistically significant correspondence with fungal loads.

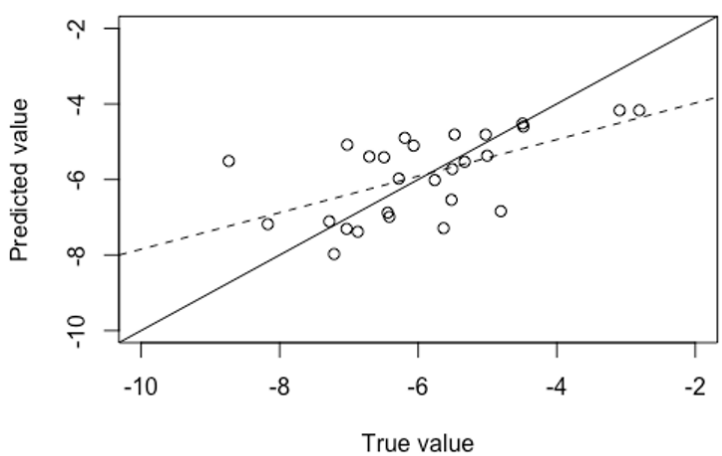

Figure 3; Comparison between actual fungal load values against predicted values from the cross-validation model. The solid line indicates the 1-to-1 line, and the dotted line shows the regression of true value on predicted value, which is highly significant (slope =0.48, $p < 0.01$).

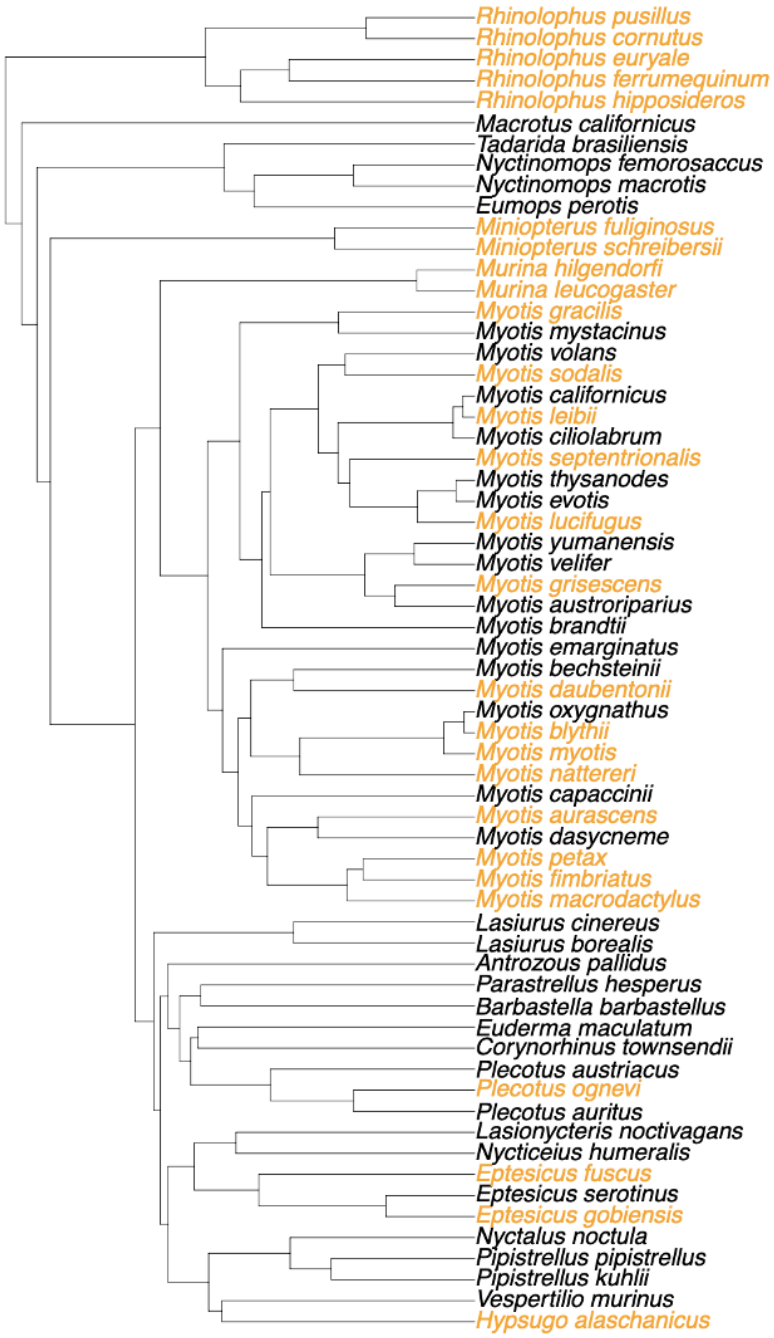

Figure 4; Phylogeny showing species with available data (Orange) and other species that will be predicted, including many from the Western USA that has yet to be exposed to WNS. Note that the species with available data are spread out on the phylogeny and hence more predictive capabilities of our model.

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